DAY-07

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Working on: "Potential questions that can be asked from version 02 PPT."

01 What is comparative study in this?

(A) 1.1 The study compares different approches for modeling pinhole defects in GS TSV'S. 02

1.2 In this work we will model GS TSV pinhole defect as RLC model and provide more comprehensive electrical characterization 03 as well as performance analysis.

1.3 Then we will compare this RLC model electrical charaterization with previous model like RL & CG_{leak} model.

02 What are GS TSV's?

(A) G S G

G → are connect to ground and placed around S.

S → carries data single

G — it acts like sound proof
- → Reduces noise
- → Reduce crosstalk
- → improve signal Integrity.
- → Improve ↑freq" performance.

03 What is electrical characterization

(A) 3.1 It is a process of evaluating the electrical performance of TSV using or measuring diff" electrical parameters

→ S_{11}, S_{12}, Leakage current, leakage power, Noise, delay, cross talk.

3.2 Noise: unwanted electrical disturbance that changes original output. 1v → 1v ideal; 1v → 0.94 + unwanked

crosstalk:

→ signal in one TSV interrupts the signal of other TSV.

→ In this what comparative study we are doing

Previous: RL, C_{leak}, G_{leak} — pinhole defect model

compare

Now: RLC — pinhole defect model

↓ it terms of electrical characterization.

04 How via last is connected to what we are designing [RLC]

→ Via F, M, L

→ why choose last

→ How via last is connected to this RLC modelling of pinhole defect.

05 Fabrication

wafer selection — orientation [miller dices] + thickness

↓

Lithograph — get via location

↓

DRIE — Via formation: itchs with very high aspect ration [depth : width]

↓

Ins

↓

Barrier → seed layer → Cu filling.

02 Working on "Equations"

02 Inductance
01 Self [If one TSV present]
02 Mutual [If two or more TSV are present]

03 Capacitance
01 7 types of capacitance.

TSV defect free

01 why we are choosing a cylindrical TSV.

04 An defect circuit diagram
01 Modelling & Simulation Reference paper.
02 Multitone dither.

06: Types of Networks & models.

05 why choosing π

01 Why we are choosing cylindrical TSV Before moving towards equations?

A
01 Easier fabrication: DRIE (Deep Reactive Ion Etching) naturally. produces nearly circular holes.

02 Uniform Electric field Distribution: field lines spread symmetrical. around TSV & Reduces the field concentration at edges.

03 Lower stress concentration. Since a cylinder have sharp corners which reduces ~~Stress~~ Electric stress.

02 Inductance (TSV)

2.1 Self inductance.

Single TSV → RLGC
Multiple TSV → 1 RLGC
2 MI
3 MC

Inductance

self	mutual.
(single TSV)	(Two TSVs)

1 Self (general equation of L of cylinder)

length of cylinder

$$L = \frac{\mu l}{2\pi} \ln\left[\frac{b}{a}\right]$$

b → reference distance. return path & goes path distance.
a → radius.

→ For an isolated cylinder

$$L = \frac{\mu_0 l}{2\pi}\left[\ln\left(\frac{2l}{r}\right) - 1\right]$$

$$B(r) = \frac{\mu I}{2\pi r}$$ → magnetic field

⇒ $\int B(r)\,dr$ → inductance (stored M field)

✱ ln → tells us how much space is available for magnetic field to spread around.

$$\boxed{L_{TSV} = \frac{1}{2}\left[\frac{\mu_0 \mu_{rTSV} (h_{TSV})}{2\pi} \ln\left[\frac{P_{TSV}}{r_{TSV}}\right]\right]}$$

because I ↑ signal ↓ ground return
So only half.

2 Mutual ~~capac~~ Inductance.

TSV MuI TSV

↳ Mutual

$$= \frac{\mu_0 \mu_{r_{TSV}}}{2\pi}\left[\ln\left[\frac{2 l_{TSV}}{s_{TSV}}\right] - 1\right]$$

→ general

general

$$L = \frac{\mu l}{2\pi}\left[\ln\left(\frac{2l}{s}\right) - 1\right]$$

l → length of cyllinder.

s → ~~radius of~~ centre to center between two TSV.

$$\boxed{L_M = \frac{\mu_0 \mu_{r_{TSV}}}{2\pi}\left[\ln\left[\frac{2 l_{TSV}}{s_{TSV}}\right] - 1\right]}$$

L signal

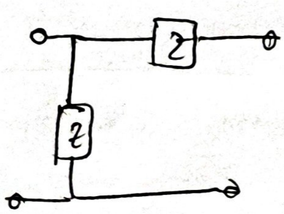
Capacitan formula.

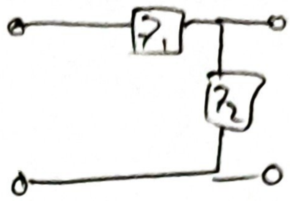
Pi
T
L } Network

lumped
distributed } models.

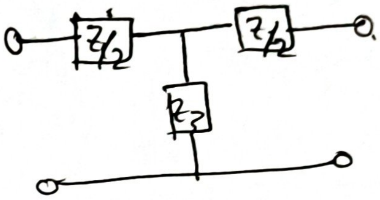
Types of Networks.

L network:-

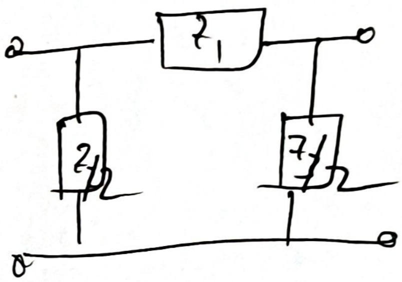




T-Network.



π Network



Types of ~~electrical~~ models [Electrical]

01 lumped model

02 distributed model.

[Elements]

lumped → which are seperable

Ex:- Resistors, Inductors, Capacitor.

distributed → which are un-seperable

Ex:- Transmission line (RLC model)